A.K.T.U PYQs

Unit Wise – Marks Wise

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||| COMPUTER ORGANIZATION AND ARCHITECTURE (BCS352) |||

|Unit-1|
|| Introduction ||

2-Marks Questions

- **[2023-24]** What are the different types of Buses used in computer architecture?
- **[2024-25]** Enlist the basic functional units of a digital system?
- **[2024-25]** Define general-purpose registers in a processor?
- **[2022-23]** How memory read and write operations are performed in computer system?
- **[2022-23]** Define bus and memory transfer?
- **[2021-22]** List and briefly define the main structural components of a computer.
- **[2021-22]** Represent the following conditional control statements by two register transfer statements with control functions. If(P=1) then (R1 <- R2) else if (Q=1) then (R1 <- R3)
- **[2021-22]** Differentiate between Daisy chaining and centralized parallel arbitration.
- **[2020-21]** Define the term Computer architecture and Computer organization.
- **[2020-21]** What is mean by bus arbitration? List different types of bus arbitration.

7-Marks Questions

- **[2023-24]** What is meant by the term BUS arbitration? Why is it needed? How can bus arbitration be implemented in Daisy changing scheme?

- **[2023-24]** What do you mean by processor organization? Explain various types of processor organization with suitable example.
- **[2023-24]** Differentiate between Memory stack and register stack.
- **[2024-25]** Discuss the process of data transfer between registers, buses, and memory units in a digital system. Include explanations of read and write operations?
- **[2024-25]** Compare and contrast the characteristics and functionalities of address buses, data buses, and control buses?
- **[2022-23]** Explain functional units of computer system in detail.
- **[2022-23]** Explain about stack organization used in processors. What do you understand by register stack?
- **[2022-23]** What is an effective address? How it is calculated in different types of addressing modes? Explain.
- **[2021-22]** A digital computer has a common bus system for 8 registers of 16 bit each. The bus is constructed using multiplexers.

  I. How many select input are there in each multiplexer?

  II. What is the size of multiplexers needed?

  III. How many multiplexers are there in the bus?
- **[2021-22]** Draw a diagram of a Bus system in which it uses 3 state buffers and a decoder instead of the multiplexers.
- **[2021-22]** Explain in detail multiple bus organization with the help of a diagram.
- **[2021-22]** Write a program to evaluate arithmetic expression using stack organized computer with 0-address instructions. $X = (A-B) * (((C - D * E) / F) / G)$
- **[2020-21]** i. Draw a diagram of bus system using MUX which has four registers of size 4 bits each.
  ii. Evaluate the arithmetic statement $X = A + B * [C * D + E * (F + G)]$ using a stack organized computer with zero address operation instructions.
- **[2020-21]** An instruction is stored at location 400 with its address field at location 401. The address field has the value 500. A processor register R1 contains the number 200. Evaluate the effective address if the addressing mode of the instruction is (i) direct (ii) immediate (iii) relative (iv) register indirect (v) index with R1 as index register.
- **[2020-21]** What do you mean by processor organization? Explain various types of processor organization.

|Unit-2|
||Arithmetic and logic unit||

2-Marks Questions

- **[2023-24]** Name the different types of multipliers.
- **[2024-25]** Explain floating point representation?
- **[2024-25]** Discuss the role of ALU in a processor?
- **[2021-22]** Design a 4-bit combinational incremental circuit using four full adder circuits.
- **[2021-22]** Register A holds the binary values 10011101. What is the register value after arithmetic shift right? Starting from the initial number 10011101, determine the register value after arithmetic shift left, and state whether there is an overflow.
- **[2020-21]** Discuss biasing with reference to floating point representation.
- **[2020-21]** What is restoring method in division algorithm?

7-Marks Questions

- **[2023-24]** Show the multiplication process using Booth's algorithm when the following numbers are multiplied: (-12) * (-18).
- **[2023-24]** Explain in detail the principle of carry looks ahead adder and design 4-bit CLA adder.
- **[2023-24]** Represent the following decimal number in IEEE standard floating-point format in a single precision method (32 bit) representation method. (i) (85.125)10 (ii) (-307.1875)10
- **[2024-25]** Describe Booth's algorithm for signed binary multiplication. Explain how it optimizes the multiplication process and discuss its advantages over traditional methods?
- **[2024-25]** Explain the concept of look-ahead carry adders in digital arithmetic circuits. Compare and contrast their performance with conventional ripple carry adders, highlighting advantages and disadvantages?
- **[2024-25]** Discuss the process of signed operand multiplication in digital arithmetic circuits?
- **[2022-23]** Explain IEEE-754 standard for floating point representation. Express (314.175)10 in all the IEEE-754 models.
- **[2022-23]** Describe the derivation procedure of look ahead carry adder by an example with the help of block diagram.

- **[2022-23]** Show the systematic multiplication process of (-15) × (-16) using Booth's Algorithm.
- **[2021-22]** Explain 2-bit by 2-bit Array multiplier. Draw the flowchart for divide operation of two numbers in signed magnitude form.
- **[2021-22]** A binary floating-point number has seven bits for a biased exponent. The constant used for the bias is 64.

  I. List the biased representation of all exponents from -64 to +63.

  II. Show that after addition of two biased exponents, it is necessary to subtract 64 in order to have a biased exponent's sum.

  III. Show that after subtraction of two biased exponents, it is necessary to add 64 in order to have a biased exponent's difference.
- **[2021-22]** Show the multiplication process using Booth algorithm, when the following binary numbers, (+13) x (-15) are multiplied.
- **[2020-21]** Explain in detail the principle of carry look ahead adder and design 4-bit CLA adder.
- **[2020-21]** Show the systemic multiplication process of (20) X (-19) using Booth's algorithm.
- **[2020-21]** Explain IEEE standard for floating point representation. Represent the number (-1460.125)10 in single precision and double precision format.

|Unit-3|
||Control Unit||

2-Marks Questions

- **[2023-24]** What are the different phases of an instruction cycle?
- **[2023-24]** How does control unit of a computer works?
- **[2024-25]** Discuss the micro operations in computer architecture?
- **[2022-23]** List the steps involved in an instruction cycle.
- **[2022-23]** Define instruction cycle.
- **[2022-23]** Differentiate between RISC and CISC.
- **[2021-22]** Differentiate between horizontal and vertical microprogramming.
- **[2021-22]** What are the different types of instruction formats?
- **[2020-21]** Define micro operation and micro code.
- **[2020-21]** Write short note on RISC.

7-Marks Questions

- **[2023-24]** What is pipelining? What are the different stages of pipelining? Explain in detail.
- **[2023-24]** Explain the different cycles of an instruction execution.
- **[2023-24]** Differentiate between hardwired and micro programmed control unit. Explain each component of hardwired control unit organization.
- **[2024-25]** Describe the concept of horizontal and vertical microprogramming in control unit design?
- **[2024-25]** Provide a detailed overview of processor organization, covering topics such as instruction fetch, decode, execute cycles, pipelining, and parallel processing techniques?
- **[2024-25]** Describe the concept of horizontal and vertical microprogramming in control unit design? (Note: The Hindi text in this question incorrectly refers to Look Ahead Carry Adders, but the English text asks about Microprogramming).
- **[2024-25]** Explain the concept of microprogram sequencing and how it enables the execution of complex instructions through a sequence of microinstructions?
- **[2022-23]** Explain the concept of pipelining and also explain types of pipelining.

- **[2022-23]** Write a program to evaluate the arithmetic statement P = ((X - Y + Z) * (A ^ B))/(C ^ D * E) By using (i) Two address instructions (ii) One address instructions (iii) Zero address instructions.
- **[2022-23]** What are the differences between hardwired and micro-programmed control unit?
- **[2021-22]** Explain with neat diagram, the address selection for control memory.
- **[2021-22]** List the differences between hardwired and micro programmed control in tabular format. Write the sequence of control steps for the following instruction for single bus architecture. R1 <- R2 * (R3)
- **[2020-21]** Draw the flowchart for instruction cycle with neat diagram and explain.
- **[2020-21]** What is a micro program sequencer? With block diagram, explain the working of micro program sequencer.
- **[2020-21]** Differentiate between hardwired and micro programmed control unit. Explain each component of hardwired control unit organization.

|Unit-4|
||Memory||

2-Marks Questions

- **[2023-24]** Write a short note on locality of reference.
- **[2023-24]** Define 2 ½ D memory organization.
- **[2024-25]** Explain the purpose of ROM memories in computer systems?
- **[2022-23]** Define HIT and MISS ratio in memory with an example.
- **[2022-23]** List the difference between static RAM and dynamic RAM.
- **[2022-23]** Define Virtual memory.
- **[2021-22]** What is the transfer rate of an eight-track magnetic tape whose speed is 120 inches per second and whose density is 1600 bits per inch?
- **[2021-22]** What is an Associative memory? What are its advantages and disadvantages?
- **[2021-22]** Differentiate between static RAM and Dynamic RAM.
- **[2020-21]** Define hit ratio.
- **[2020-21]** What do you mean by page fault?

7-Marks Questions

- **[2023-24]** Give classification of memory based on the method of access. Also discuss construction and working of magnetic disk and various components of disk access time.
- **[2023-24]** Consider a cache (M1) and memory (M2) hierarchy with following characteristics: M1: 16K word, 50 ns Access time; M2: 1M word, 400 ns Access time. Assume 8-word cache blocks and set size 256 words with set associative mapping.
- (i) Show and explain the mapping between M2 and M1.
- (ii) Calculate the effective memory access time with cache hit ratio=0.95.
- **[2023-24]** Explain the direct mapping technique. Consider a digital computer has a memory unit of 64k*16 and cache memory of 1k words. The cache uses direct mapping with 4 block size of four words.
- (i) How many bits are there in the tag, block and word fields of the address format?
- (ii) How many blocks can the cache accommodate?

- **[2024-25]** Discuss the characteristics and functionalities of auxiliary memories such as magnetic disks, magnetic tapes, and optical disks?
- **[2024-25]** Design and explain the fundamental concept of computer memory hierarchy?
- **[2024-25]** Describe the architecture and operation of semiconductor RAM (Random Access Memory) memories such as DRAM (Dynamic RAM) and SRAM (Static RAM)?
- **[2022-23]** Consider a cache consisting of 256 blocks of 16 words each for a total of 4096 words and assume that the main memory is addressable by a 16 bits address and it consists of 4K blocks. How many bits are there in each of TAG, SET, WORD field for 2-way set associative technique?
- **[2022-23]** Discuss the Memory Hierarchy in computer system with regard to Speed, Size and Cost.
- **[2022-23]** Write a short notes on magnetic disk, magnetic tape and optical disk.
- **[2021-22]** A digital computer has a memory unit of 64K X 16 and a cache memory of 1K words. The cache uses direct mapping with a block size of four words.

  I. How many bits are there in the tag, index, block, and word fields of the address format?

  II. How many bits are there in each word of cache, and how they are divided into functions? Include a valid bit.

  III. How many blocks can the cache accommodate?
- **[2021-22]** The logical address space in a computer system consists of 128 segments. Each segment can have up to 32 pages of 4K words each. Physical memory consists of 4K blocks of 4K words each. Formulate the logical and physical address formats.
- **[2021-22]** How is the Virtual address mapped into physical address? What are the different methods of writing into cache?
- **[2020-21]** Discuss 2 D RAM and 2.5D RAM with suitable diagram.
- **[2020-21]** Calculate the page fault for a given string with the help of LRU & FIFO page replacement algorithm, Size of frames = 4 and string 1 2 3 4 2 1 5 6 2 1 2 3 7 6 3 2 1 2 3 6.
- **[2020-21]** A computer uses RAM chips of 1024*1 capacity.

  i) How many chips are needed & how should their address lines be connected to provide a memory capacity of 1024*8?

  ii) How many chips are needed to provide a memory capacity of 16 KB?

|Unit-5|
||Input / Output||

2-Marks Questions

- **[2023-24]** In what way synchronous and asynchronous serial modes of data transfer differ?
- **[2024-25]** Enlist the peripheral devices in a computer system?
- **[2022-23]** List down the functions performed by an Input/Output unit.
- **[2022-23]** Why does the DMA get priority over CPU when both request memory transfer?
- **[2020-21]** Explain the term cycle stealing.
- **[2020-21]** What do you mean by vector interrupt? Explain.

7-Marks Questions

- **[2023-24]** What are the basic differences between interrupt initiated I/O and programmed I/O? Explain in detail.
- **[2023-24]** What do you mean by asynchronous data transfer? Explain strobe control and handshaking mechanism.
- **[2023-24]** Explain the various modes of data transfer and discuss direct memory access mode in detail. Also explain how DMA is superior to other modes.
- **[2024-25]** Discuss the functions and characteristics of I/O ports and their significance in facilitating data exchange between the CPU and external devices?
- **[2024-25]** Describe the role of peripheral devices in a computer system and explain how they interact with the CPU through I/O interfaces and ports?
- **[2024-25]** Explain the Direct Memory Access (DMA) techniques for data transfer?
- **[2022-23]** Define interrupt. Also discuss different types of interrupt.
- **[2022-23]** With a neat schematic diagram, explain about DMA controller and its mode of data transfer.
- **[2022-23]** Discuss the design of a typical input or output interface.
- **[2021-22]** Explain destination-initiated transfer using handshaking method.
- **[2021-22]** Explain how the computer buses can be used to communicate with memory and I/O. Also draw the block diagram for CPU-IOP communication.
- **[2021-22]** What are the different methods of asynchronous data transfer? Explain in detail.

- **[2020-21]** Draw and explain the block diagram of typical DMA controller.
- **[2020-21]** What do you mean by asynchronous data transfer? Explain strobe control and hand shaking mechanism.
- **[2020-21]** Discuss the different modes of data transfer.